

Aryan Rajpurkar

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TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL

Web & APIs: Flask, FastAPI, React, Next.js, Spring Boot, RESTful APIs, WebSockets

Cloud & Data: AWS (EC2, S3, ECS, ECR, RDS, CloudWatch, Glue, Athena), GCP (Cloud Run), Apache (Kafka, Spark, Airflow)

DevOps: Docker, Kubernetes, Git, CI/CD, Microservices

AI/ML: TensorFlow, PyTorch, NLP, LangChain, LangGraph, CrewAI, AutoGen, RAG, FAISS, Pinecone, ChromaDB, Neo4j, Federated Learning, Generative AI, SHAP, LIME, Transformers, MCP, PowerBI

Platforms: HeyGen, Arcads.ai, n8n, OpenCV, Supabase, Clerk API, Astra DB

PROFESSIONAL EXPERIENCE

Founder

Aretis Labs | aretislabs.com

India

Jan 2026 – Present

- Founded an **AI automation consultancy** building autonomous, human-in-loop systems for enterprise clients; scaled to **10+ enterprise clients** within 5 months.
- Delivered AI-native marketplace and workflow automation for **Polycab** and **Aerem Solar**: autonomous agents handling procurement decisions, supplier matching, and exception routing with full audit trails.
- Architected **RAG-based enterprise knowledge engines** (private, on-premise) with source attribution on every answer, enabling clients to query internal docs, contracts, and communications in natural language.
- Built **Generative Engine Optimization (GEO)** pipelines to improve clients' brand citation rates across LLMs (ChatGPT, Gemini, Perplexity), tracking share-of-voice across AI models as a measurable KPI.
- **Tech:** Python, LangGraph, CrewAI, FastAPI, Next.js, PostgreSQL, Supabase, RAG, n8n, AWS

Founding Engineer, Data Platforms

VisaFriendly | visafriendly.com

India (Remote)

June 2025 – Present

- Built the **end-to-end data platform from scratch**: designed cloud infrastructure on **AWS ECS, CloudWatch, S3, RDS**; architected ETL pipelines processing **100K+ monthly job entries** with data quality contracts and lineage tracking across ingestion, transformation, and storage, improving downstream accuracy by **28%** and cutting processing time by **40%**.
- Designed **multi-agent pipeline** (CrewAI + n8n) with structured tool definitions and output evaluation against rubric, automating **8K documents/month** and improving interview response rates by **22%**.
- Built an **AI-native UGC content generation pipeline**: agents pull context from live job-market news feeds and industry developments, generate platform-native scripts, render AI avatar video creators via **Arcads.ai**, and auto-schedule posts to Instagram, LinkedIn, and Twitter, making the entire marketing pipeline fully autonomous.
- Exposed ETL pipeline as an **MCP server**, enabling Claude/Cursor to query the job data catalog and pipeline metadata in natural language, reducing ad-hoc data debugging time significantly.
- **Tech:** AWS (ECS, ECR, CloudWatch, S3, RDS), Python, SQL, CrewAI, n8n, Arcads.ai, MCP, ETL, LLMs

Founding Engineer, AI

Mantrika.ai | mantrika.ai

India

Jan 2025 – Aug 2025

- Joined as founding engineer to build the core AI-powered HR recruitment platform from zero; owned **end-to-end architecture**, database schema design, and all agentic infrastructure.
- Integrated **HeyGen** to build an AI avatar-based interview system: candidates complete voice and video AI interviews with a virtual interviewer, with responses evaluated automatically against structured rubrics.
- Built a **computer vision proctoring system** using **OpenCV**: real-time eye-tracking detects gaze deviation and focus loss during candidate assessments, with anomaly flagging and a tamper-resistant audit log per session.
- Architected agentic recruitment workflows using **LangGraph state machines** with fallback routing across LLM providers, achieving **40% latency improvement** and **20% reduction** in cost per candidate selection.
- **Tech:** TypeScript, Next.js, LangGraph, LangChain, MySQL, Supabase, AWS, CrewAI, HeyGen, OpenCV

NLP (ML) Engineer Intern

AI4Education Labs | ai4educationlabs.com

Germany (Remote)

June 2024 – July 2024

- Built an AI-driven adaptive assessment platform combining **Item Response Theory (IRT)** with **knowledge graphs** to model per-student skill gaps and dynamically adjust question difficulty; impacted **5,000+ students** across 10+ schools in India and Germany.
- Engineered an end-to-end **NLG pipeline** for personalized content recommendations: entity extraction, concept graph traversal, and template-grounded generation achieving **95% accuracy**; deployed on **AWS EC2** with assets stored on **S3**.
- **Tech:** Python, NLP, Knowledge Graphs, Item Response Theory, AWS (S3, EC2), NLG, Transformers

- Deployed **multimodal AI agents** on Telegram, WhatsApp, and Instagram for FMCG retail audit automation: agents parsed image inputs, executed structured tool calls against audit rubrics, and routed exceptions with fallback logic, reducing audit turnaround time by **35%**.
- Built an **Influencer Marketing platform** using Instagram Graph API and LangChain: semantic brand-fit scoring via embedding similarity, engagement analytics ingestion via **GCP BigQuery**, and creator-brand matching pipeline deployed on **Cloud Run**.
- Engineered **prompt pipelines** with structured output schemas and retry logic to ensure consistent JSON responses from LLMs across variable input formats and languages.
- **Tech:** Python, LangChain, Generative AI, GCP (BigQuery, Cloud Run), SQL, Prompt Engineering, REST APIs

PROJECTS

Launchy: AI-Native Content Studio | launchy.aryanrajpurkar.com | *Live Demo*

- AI-native studio for creators and marketers: **6 CrewAI LLM agents** with Reddit + web search grounding; citation-style provenance (outputs tied to specific sources); **retrieval quality gates** and abstention logic when context is insufficient to generate trustworthy content.
- 5×2 default angle/variation matrix (up to 10×10) with one-click publish to Twitter, LinkedIn, and Instagram; **DAG introspection** tracing signal → agent → artifact for full run observability.
- 10 content blocks / 9 governed templates with **versioned prompts & JSON schemas**; streamed FastAPI runs with per-node artifacts; **Chroma RAG** (top-k 5–20), 0–100 relevance scoring, **predicted_score** refresh via CSV ingest.
- **Tech:** Python, CrewAI, ChromaDB, FastAPI, TypeScript, React, Vite, WebSockets, Serper API

Equitas: Enterprise AI Reliability & Safety Platform | equitas.aryanrajpurkar.com | *PyPI Package*

- Enterprise AI evaluation and safety layer with **multi-layer detection** (toxicity, bias, hallucination, jailbreak) backed by a **fine tuned toxicBERT** running **500+ adversarial test cases**; designed as a drop-in API replacement so teams can trust AI outputs on sensitive enterprise data.
- **SHAP/LIME explainability** exposed via REST API for enterprise audit trails, with every flagged output including a traceable explanation; multi-tenant architecture with per-tenant guardrail configs.
- **Tech:** Python (FastAPI), React/TypeScript, MongoDB, PyTorch, Transformers, SHAP, LIME, Redis, Docker, LLMs, ExplainableAI (X-AI)

Sahayak AI : Metadata-First AI Document Operations Platform | sahayak-ai.aryanrajpurkar.com

- Document intelligence platform with **precedent relationship graph** modeling provenance, dependencies, and **citation lineage across government documents**, enabling graph-traversal queries over document metadata rather than just text.
- Implemented **hybrid RAG** with **BM25 + FAISS vector retrieval**, **cross-encoder reranking**, machine-readable compliance checks and automated workflow routing via Apache Kafka.
- Integrated **MCP and published catalog operations as agent-callable tools** - hybrid search, metadata fetch, contradiction analysis, version chains, and grounded document Q&A—for IDE / LLM clients.
- **Tech:** Next.js, Python (Flask), MongoDB (GridFS), Neo4j, BM25, FAISS, Elasticsearch, OCR, Apache Kafka

LEADERSHIP & AWARDS

- **Chairperson, DJS-S4DS**, National Data Science Committee of India; mentored **200+ students** in ML/AI
- **Winner, Smart India Hackathon 2024**, National hackathon by the Government of India
- **Amazon ML Summer School Scholar 2024**, selected from 61,000 participants nationwide
- **Best Student Chair 2025**, Society of Data Science, India
- **5× National Hackathon Winner**

EDUCATION

D.J. Sanghvi College of Engineering

B.Tech in Computer Science and Engineering (Data Science), CGPA: 9.34/10.0

Mumbai, India

2022 – 2026